

3GPP TS 31.130 V16.0.0 (2020-07)

Technical Specification

3rd Generation Partnership Project; Technical Specification Group Core Network and Terminals; (U)SIM Application Programming Interface (API); (U)SIM API for Java™ Card; (Release 16)



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3GPP

Postal address

3GPP support office address

650 Route des Lucioles - Sophia Antipolis
Valbonne - FRANCE
Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Internet

<http://www.3gpp.org>

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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

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where:

- x the first digit:
 - 1 presented to TSG for information;
 - 2 presented to TSG for approval;
 - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

- shall** indicates a mandatory requirement to do something
- shall not** indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

- should** indicates a recommendation to do something
- should not** indicates a recommendation not to do something
- may** indicates permission to do something
- need not** indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

- can** indicates that something is possible
- cannot** indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

- will** indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document
- will not** indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document
- might** indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document defines the (U)SIM Application Programming Interface extending the "UICC API for Java Card™" [2].

This API allows to develop a (U)SAT application running together with a (U)SIM application and using 3GPP network features.

The present document includes information applicable to 3GPP network operators, service providers, server – (U)SIM – database manufacturers.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

- [1] ETSI TS 101 220: "Integrated Circuit Cards (ICC); ETSI numbering system for telecommunication; Application providers (AID)".
- [2] ETSI TS 102 241 V15.0.0: "UICC API for Java Card™"
- [3] 3GPP TS 31.102: "Characteristics of the USIM Application".
- [4] 3GPP TS 51.011 Release 4: "Specification of the Subscriber Identity Module- Mobile Equipment (SIM – ME) interface".
- [5] 3GPP TS 23.041: "Technical realization of Cell Broadcast Service (CBS)".
- [6] 3GPP TS 31.101: "UICC-terminal interface; Physical and logical characteristics".
- [7] 3GPP TS 31.111: "USIM Application Toolkit (USAT)".
- [8] 3GPP TS 51.014 Release 4: "Specification of the SIM Application Toolkit for the Subscriber Identity Module – Mobile Equipment (SIM – ME) interface".
- [9] 3GPP TS 31.115: "Secured packet structure for the (U)SIM Toolkit applications".
- [10] 3GPP TS 23.040: "Technical realization of the Short Message Service (SMS)".
- [11] ORACLE "Application Programming Interface, Java Card™ Platform, 3.0.1 Classic Edition".
- [12] ORACLE "Runtime Environment Specification, Java Card™ Platform, 3.0.1 Classic Edition".

- [13] ORACLE "Virtual Machine Specification Java Card™ Platform, 3.0.1 Classic Edition".

Note: ORACLE Java Card™ Specifications can be downloaded at
<http://docs.oracle.com/javame/javacard/javacard.html>

- [14] 3GPP TS 23.032: "Universal Geographical Area Description (GAD)".

- [15] IEC 61162-1: "Maritime navigation and radio communication equipment and systems – Digital interfaces".

3 Definitions and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions defined in ETSI TS 102 241 [2] apply.

(U)SAT Framework : (U)SAT extension of the CAT Runtime Environment.

3.2 Abbreviations

For the purposes of the present document, the abbreviations defined in ETSI TS 102 241 [2] apply.

4 Description

4.0 Overview

This API is an extension to the ETSI TS 102 241 [2] "UICC API for Java Card™" and requires the implementation of this specification.

The classes and interfaces described in this specification inherit functionality from the classes and interfaces specified in the "UICC API for Java Card™".

The (U)SAT Framework described in this specification is an extension of the CAT Runtime Environment defined in ETSI TS 102 241 [2].

4.1 (U)SIM Java Card™ Architecture

The overall architecture of the (U)SIM API is based on the "UICC API for Java Card™" defined in ETSI TS 102 241 [2].

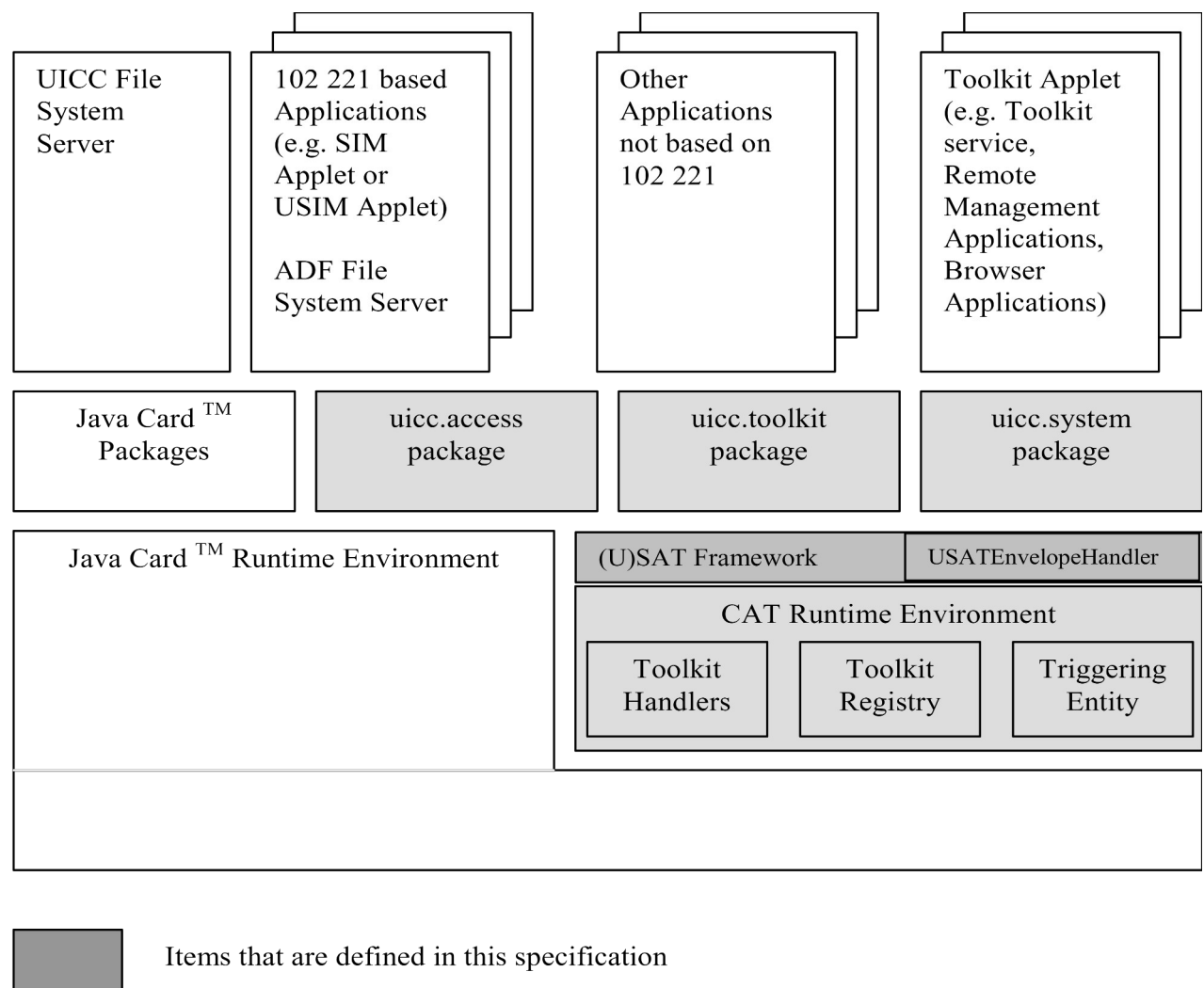


Figure 1: (U)SIM Java Card™ Architecture

5 File Access API

The (U)SIM file access API consists of the package *uicc.usim.access*. This package defines additional constants to those defined in the *uicc.access* package from ETSI TS 102 241 [2]. The access to the file system, defined in TS 51.011 [4] and TS 31.102 [3], is the one specified in ETSI TS 102 241 [2] via the UICC *FileView* Interface. When selecting a cyclic file the current record number is defined, this applies also to files located under DF_{GSM}.

6 (U)SAT Framework

6.0 Overview

The (U)SIM toolkit API consists of the *uicc.usim.toolkit* package for toolkit features defined in TS 31.111 [7] and TS 51.014 [8], and is based on the *uicc.toolkit* package defined in ETSI TS 102 241 [2].

6.1 Applet triggering

See ETSI TS 102 241 [2].

6.1.1 Exception Handling

The following clause describes the handling of exceptions by the (U)SAT Framework in addition to the behaviour defined in ETSI TS 102 241 [2] for the CAT Runtime Environment.

If an Applet triggered by `EVENT_FORMATTED_SMS_PP_ENV` event throws an `ISOException` with the reason code (0x6FXX), it shall be sent to the terminal.

Other Exceptions shall not be propagated to the terminal.

6.2 Definition of Events

The following events can trigger a Toolkit Applet in addition to the events defined in ETSI TS 102 241 [2], all short values are reserved in ETSI TS 102 241 [2]:

Table 1: (U)SAT event list

Event Name	Reserved short value
<code>EVENT_FORMATTED_SMS_PP_ENV</code>	2
<code>EVENT_FORMATTED_SMS_PP_UPD</code>	3
<code>EVENT_UNFORMATTED_SMS_PP_ENV</code>	4
<code>EVENT_UNFORMATTED_SMS_PP_UPD</code>	5
<code>EVENT_UNFORMATTED_SMS_CB</code>	6
<code>EVENT_MO_SHORT_MESSAGE_CONTROL_BY_NAA</code>	10
<code>EVENT_FORMATTED_SMS_CB</code>	24
<code>EVENT_EVENT_DOWNLOAD_IWLAN_ACCESS_STATUS</code>	30
<code>EVENT_EVENT_DOWNLOAD_NETWORK_REJECTION</code>	31
<code>EVENT_EVENT_DOWNLOAD_CSG_CELL_SELECTION</code>	33
<code>EVENT_EVENT_DOWNLOAD_DATA_CONNECTION_STATUS_CHANGE</code>	37
<code>EVENT_FORMATTED_USSD</code>	121
<code>EVENT_UNFORMATTED_USSD</code>	122
<code>EVENT_EVENT_DOWNLOAD_IMS_REGISTRATION</code>	119
<code>EVENT_EVENT_DOWNLOAD_INCOMING_IMS_DATA</code>	120

EVENT_FORMATTED_SMS_PP_ENV, EVENT_UNFORMATTED_SMS_PP_ENV, EVENT_FORMATTED_SMS_PP_UPD, EVENT_UNFORMATTED_SMS_PP_UPD

There are two ways for a card to receive a Short Message Point to Point: via an `ENVELOPE(SMS-PP DOWNLOAD)` APDU as defined in TS 31.111 [7] and TS 51.014 [8] or an `UPDATE RECORD EFSMS` APDU as defined in TS 31.102 [3] and TS 51.011 [4]. The `EFSMS` can be either located under the `DFTelecom` or under any ADF as defined in TS 31.102 [3] and TS 51.011 [4].

The received Short Message may be:

- formatted according to TS 31.115 [9] or an other protocol to identify explicitly the toolkit applet for which the message is sent;
- unformatted (e.g. a toolkit applet specific protocol) then the (U)SAT Framework will pass this data to all registered toolkit applets.

When the Short Message is received as Concatenated Short Messages as defined in TS 23.040 [10], it is the responsibility of the (U)SAT Framework to link single Short Messages together to re – assemble the original message before any further processing. The original Short Message shall be placed in one SMS TPDU TLV (with TP-UDL field coded on one octet) included in the *USATEnvelopeHandler*. The concatenation control headers used to re-assemble the short messages in the correct order shall not be present in the SMS TPDU. The TP-elements of the SMS TPDU and the Address (TS – Service-Centre-Address) shall correspond to the ones in the last received Short Message (independently of the Sequence number of Information-Element-Data).

The minimum requirement for the (U)SAT Framework is to process a concatenated short message with the following properties:

- the Information Element Identifier is equal to the 8-bit reference number.
- it contains uncompressed 8 bit data or uncompressed UCS2 data.

EVENT_FORMATTED_SMS_PP_ENV

Upon reception of a TS 31.115 [9] formatted Short Message Point to Point (Single or Concatenated) via an ENVELOPE, the (U)SAT Framework shall:

- verify the security of the Short Message as per TS 31.115 [9];
- trigger the toolkit applet registered with the corresponding TAR;
- take the optional Application Data posted by the triggered toolkit applet if present;
- secure and send the response packet using SMS-DELIVER-REPORT or SMS-SUBMIT.

When the toolkit applet is triggered, data shall be provided deciphered.

EVENT_UNFORMATTED_SMS_PP_ENV

Upon reception of an unformatted Short Message Point to Point (Single or Concatenated) via an ENVELOPE, the (U)SAT Framework shall trigger all the Toolkit Applets registered to this event.

NOTE: As a consequence of the *EnvelopeResponseHandler* availability rules specified in clause 6.6, only the first triggered toolkit applet is guaranteed to be able to send back a response.

EVENT_FORMATTED_SMS_PP_UPD

Upon reception of a TS 31.115 [9] formatted Short Message Point to Point (Single or Concatenated) via an UPDATE RECORD EF_{SMS}, the (U)SAT Framework shall:

- update the EF_{SMS} file with the data received, it is then up to the receiving toolkit applet to change the SMS stored in the file (i.e. the toolkit applet need to have access to the EF_{SMS} file)
- verify the security of the Short Message as per TS 31.115 [9];
- convert the UPDATE RECORD EF_{SMS} APDU into a COMPREHENSION TLV List;
- trigger the toolkit applet registered with the corresponding TAR;

When the toolkit applet is triggered, data shall be provided deciphered.

The *USATEnvelopeHandler* provided to the applet shall:

- return *BTAG_SMS_PP_DOWNLOAD* to the *getTag()* method call;
- return the Comprehension TLV list length to the *getLength()* method call;

The *USATEnvelopeHandler* provided to the applet shall contain the following COMPREHENSION TLVs:

- Device Identities TLV

The Device Identities Comprehension TLV is used to store the information about the absolute record number in the EF_{SMS} file and the value of the EF_{SMS} record status byte, and is formatted as defined below:

Device identities Comprehension TLV
Device Identities tag
length = 02
Absolute Record Number
Record Status

With the absolute record number the toolkit applet can update EF_{SMS} in absolute mode to change the received SMS (e.g. in a readable text).

For Concatenated Short Message the Absolute Record Number and the Record Status will correspond to the last UPDATE RECORD EF_{SMS} APDU received.

- Address TLV

The value is the TS-Service-Centre-Address (RP-OA) of the last UPDATE RECORD EF_{SMS} APDU.

- SMS TPDU TLV

The value is the SMS TPDU provided deciphered and reassembled, if needed

- AID TLV

The AID comprehension TLV is present only if the EF_{SMS} file updated is under an ADF. The value is the AID of the ADF as defined TS 31.111 [7].

The order of the TLVs given in the *USATEnvelopeHandler* is not specified,

NOTE: To get each COMPREHENSION TLV, it is recommended that the applet uses the *ViewHandler.findTLV()* methods

EVENT_UNFORMATTED_SMS_PP_UPD

Upon reception of an unformatted Short Message Point to Point (Single or Concatenated) via UPDATE RECORD EF_{SMS} APDU, the (U)SAT Framework shall :

- update the EF_{SMS} file with the data received;
- convert the UPDATE RECORD EF_{SMS} APDU data into a COMPREHENSION TLV List (as described for *EVENT_FORMATTED_SMS_PP_UPD*);
- trigger all the Toolkit Applets registered to this event.

The content of EF_{SMS} may have been modified by a previously triggered Toolkit Applet..

EVENT_FORMATTED_SMS_CB, EVENT_UNFORMATTED_SMS_CB

The received Cell Broadcast Message, via an ENVELOPE (CELL BROADCAST DOWNLOAD) APDU as defined in TS 31.111 [7] and TS 51.014 [8] and, can be either:

- formatted according to TS 31.115 [9] or an other protocol to identify explicitly the toolkit applet for which the message is sent;
- unformatted (e.g. using a toolkit applet specific protocol), then the (U)SAT Framework will pass this data to all registered toolkit applets.

When the Cell Broadcast Message is received as multiple pages as defined in TS 23.041 [5], it is the responsibility of the (U)SAT Framework to link single pages together to re-assemble the original message before any further processing. The original Cell Broadcast message shall be placed in one Cell Broadcast page TLV included in the *USATEnvelopeHandler*. The message parameters shall correspond to the ones in the last received Cell Broadcast page (independently of the Page Parameter).

EVENT_FORMATTED_SMS_CB

Upon reception of a TS 31.115 [9] formatted Cell Broadcast message, the (U)SAT Framework shall:

- verify the security of the Cell Broadcast message as per TS 31.115 [9];
- trigger the toolkit applet registered with the corresponding TAR;

When the toolkit applet is triggered, data shall be provided deciphered.

EVENT_UNFORMATTED_SMS_CB

Upon reception of an unformatted Cell Broadcast message, the (U)SAT Framework shall trigger all the Toolkit Applets registered to this event.

EVENT_MO_SHORT_MESSAGE_CONTROL_BY_NAA

Upon reception of an ENVELOPE (MO SHORT MESSAGE CONTROL defined in TS 51.014 [8] and TS 31.111 [7]) APDU as defined in TS 31.101 [6] and TS 51.011 [4] the (U)SAT Framework shall trigger the Toolkit Applet registered to this event. The (U)SAT Framework shall not allow more than one Toolkit Applet to be registered to this event at a time (e.g. if a Toolkit Applet is registered to this event but not in selectable state the (U)SAT Framework shall not allow another Toolkit Applet to register to this event).

EVENT_FORMATTED_USSD, EVENT_UNFORMATTED_USSD

The received USSD String, via an ENVELOPE (USSD Data Download) APDU as defined in TS 31.111 [7], may be:

- formatted according to TS 31.115 [9] or an other protocol to identify explicitly the toolkit applet for which the message is sent;
- unformatted (e.g. a toolkit applet specific protocol) then the (U)SAT Framework will pass this data to all registered toolkit applets.

When the USSD Message is received as concatenated as defined in TS 31.115 [9], it is the responsibility of the (U)SAT Framework to link single USSD Messages together to re-assemble the original message before any further processing. The original USSD message shall be placed in one USSD String TLV included in the *USATEnvelopeHandler*. The USSD String parameters (DCS, PFI, CCF) shall correspond to the ones in the last received USSD String (independently of the CCF Sequence number).

EVENT_FORMATTED_USSD

Upon reception of a TS 31.115 [9] formatted USSD Message via an ENVELOPE, the (U)SAT Framework shall:

- verify the security of the USSD Message as per TS 31.115 [9];
- trigger the toolkit applet registered with the corresponding TAR;
- take the optional Application Data posted by the triggered toolkit applet if present;
- secure and send the response packet.

When the toolkit applet is triggered, data shall be provided deciphered.

EVENT_UNFORMATTED_USSD

Upon reception of an unformatted USSD String via an ENVELOPE, the (U)SAT Framework shall trigger all the Toolkit Applets registered to this event.

NOTE: As a consequence of the *EnvelopeResponseHandler* availability rules specified in clause 6.6, only the first triggered toolkit applet is guaranteed to be able to send back a response.

*EVENT_EVENT_DOWNLOAD_IWLAN_ACCESS_STATUS**EVENT_EVENT_DOWNLOAD_NETWORK_REJECTION**EVENT_EVENT_DOWNLOAD_CSG_CELL_SELECTION**EVENT_EVENT_DOWNLOAD_IMS_REGISTRATION**EVENT_EVENT_DOWNLOAD_INCOMING_IMS_DATA**EVENT_EVENT_DOWNLOAD_DATA_CONNECTION_STATUS_CHANGE*

Upon reception of an ENVELOPE (Event Download) APDU command as defined in TS 31.111 [7] the (U)SAT Framework shall trigger all the Toolkit applets registered to the corresponding event.

The following events defined in TS 31.111 [7] shall be raised upon reception of the corresponding APDU defined in either TS 51.011 [4] or TS 31.101 [6].

*EVENT_PROFILE_DOWNLOAD**EVENT_MENU_SELECTION, EVENT_MENU_SELECTION_HELP_REQUEST*

EVENT_CALL_CONTROL_BY_NAA
EVENT_TIMER_EXPIRATION
EVENT_EVENT_DOWNLOAD_MT_CALL
EVENT_EVENT_DOWNLOAD_CALL_CONNECTED
EVENT_EVENT_DOWNLOAD_CALL_DISCONNECTED
EVENT_EVENT_DOWNLOAD_LOCATION_STATUS
EVENT_EVENT_DOWNLOAD_USER_ACTIVITY
EVENT_EVENT_DOWNLOAD_IDLE_SCREEN_AVAILABLE
EVENT_EVENT_DOWNLOAD_CARD_READER_STATUS
EVENT_STATUS_COMMAND
EVENT_EVENT_DOWNLOAD_LANGUAGE_SELECTION
EVENT_EVENT_DOWNLOAD_BROWSER_TERMINATION
EVENT_EVENT_DOWNLOAD_DATA_AVAILABLE
EVENT_EVENT_DOWNLOAD_CHANNEL_STATUS
EVENT_EVENT_DOWNLOAD_ACCESS_TECHNOLOGY_CHANGE
EVENT_EVENT_DOWNLOAD_DISPLAY_PARAMETER_CHANGED
EVENT_EVENT_DOWNLOAD_LOCAL_CONNECTION
EVENT_EVENT_DOWNLOAD_NETWORK_SEARCH_MODE_CHANGE
EVENT_EVENT_DOWNLOAD_BROWSING_STATUS
EVENT_PROACTIVE_HANDLER_AVAILABLE
EVENT_EXTERNAL_FILE_UPDATE
EVENT_FIRST_COMMAND_AFTER_ATR
EVENT_UNRECOGNIZED_ENVELOPE

6.3 Registration

A Toolkit Applet shall register to events described in 6.2 as defined in ETSI TS 102 241 [2].

Constants for these events are available in *uicc.usim.toolkit.ToolkitConstants* interface in Annex A.

The *uicc.toolkit.ToolkitException* `TAR_NOT_DEFINED` shall be thrown if a Toolkit Applet has no TAR defined and registers to events: `EVENT_FORMATTED_SMS_PP_ENV`, `EVENT_FORMATTED_SMS_PP_UPD`, `EVENT_FORMATTED_SMS_CB`, `EVENT_FORMATTED_USSD`.

The *uicc.toolkit.ToolkitException*.`EVENT_ALREADY_REGISTERED` shall be thrown if there is another Toolkit Applet already registered to *EVENT_MO_SHORT_MESSAGE_CONTROL_BY_NAA*.

6.4 Proactive command handling

There is no extension of the CAT Runtime Environment by the (U)SAT Framework for proactive command handling.

6.5 Envelope response handling

For the events defined in the present document, the following rules apply:

A Toolkit Applet can post a response by using the *post()* method or the *postAsBERTLV()* method defined in ETSI TS 102 241 [2]. The (U)SAT Framework shall return the Status Word as defined in TS 31.111 [7] and in TS 51.014 [8] depending on the current NAA.

Case of EVENT_MO_SHORT_MESSAGE_CONTROL_BY_NAA:

- The rules defined for *EVENT_CALL_CONTROL_BY_NAA* in ETSI TS 102 241 [2] apply.

Case of EVENT_UNFORMATTED_SMS_PP_ENV:

- See ETSI TS 102 241 [2].

Case of EVENT_FORMATTED_SMS_PP_ENV:

- When the *post()* or the *postAsBERTLV()* method is invoked, the (U)SAT Framework shall, according to bit 6 of the second octet of the SPI defined in TS 31.115 [9], build a SMS-DELIVER-REPORT or a SMS-SUBMIT.

In case of a SMS-DELIVER-REPORT and if the post response is too large to be contained in a SMS-DELIVER-REPORT, the (U)SAT Framework shall issue Response Packets as defined in TS 31.115 [9].

In case of a SMS-DELIVER-REPORT, the (U)SAT Framework shall return the Status Word for RP-ACK or RP-ERROR as defined in TS 31.111 [7] and in TS 51.014 [8] depending on the current NAA.

In case of SMS-SUBMIT the boolean value method parameter shall be ignored by the (U)SAT Framework. If the SMS-SUBMIT is to be used, the (U)SAT Framework shall build and issue a Send Short Message proactive command as defined in TS 31.111 [7] and in TS 51.014 [8] depending on the current NAA .

Case of EVENT_FORMATTED_USSD:

- When the *post()* or the *postAsBERTLV()* method is invoked, the (U)SAT Framework shall build a USSD String to be sent back in the Return Result Component contained in the subsequent Facility message. In that case the (U)SAT Framework shall return the Status Word as defined in TS 31.111 [7].

Case of EVENT_UNFORMATTED_USSD:

- See ETSI TS 102 241 [2].

6.6 System Handler management

For the handler management of the *ProactiveHandler*, the *ProactiveResponseHandler*, the *EnvelopeHandler* and the *EnvelopeResponseHandler*, the rules defined in ETSI TS 102 241 [2] apply.

USATEnvelopeHandler:

The single system instance of the *USATEnvelopeHandler* and the single system instance of the *EnvelopeHandler* are two distinct objects instances.

- When available the *USATEnvelopeHandler* shall remain available and its content shall remain unchanged from the invocation to the termination of the *processToolkit()* method.
- The TLV List provided in the *USATEnvelopeHandler* are the same as in the *EnvelopeHandler*.
- The handler availability of the *USATEnvelopeHandler* is the same handler availability as the *EnvelopeHandler* including all the events defined in ETSI TS 102 241 [2].

The following table describes the minimum availability of the handlers for all the events at the invocation of the *processToolkit()* method of the Toolkit Applet. The rules described in this table apply in addition to the rules described in "UICC API for Java Card™"

Table 2: Handler availability for each event

EVENT_	Reply busy allowed	EnvelopeHandler / USATEnvelopeHandler	EnvelopeResponse Handler	Nb of triggered / registered Applet
_FORMATTED_SMS_PP_ENV	Y (see Note 1)	Y	Y	1 / n (per TAR)
FORMATTED_SMS_PP_UPD	N	Y	N	1 / n (per TAR)
UNFORMATTED_SMS_PP_ENV	Y	Y	Y	n / n
UNFORMATTED_SMS_PP_UPD	N	Y	N	n / n
FORMATTED_SMS_CB	Y	Y	N	1/n (per TAR)
UNFORMATTED_SMS_CB	Y	Y	N	n / n
MO_SHORT_MESSAGE_CONTROL_BY_NAA	N	Y	Y	1 / 1
FORMATTED_USSD	Y	Y	Y	1 / n (per TAR)
UNFORMATTED_USSD	Y	Y	Y	n / n
EVENT_DOWNLOAD				
IWLAN_ACCESS_STATUS	Y	Y	N	n/n
NETWORK_REJECTION	Y	Y	N	n/n
IMS_REGISTRATION	Y	Y	N	n/n
INCOMING_IMS_DATA	Y	Y	N	n/n
DATA_CONNECTION_STATUS_CHANGE	Y	Y	N	n/n

NOTE 1: The framework may reply busy and not trigger the toolkit applet if e.g. a PoR using SMS SUBMIT is required in the incoming message and a proactive session is ongoing.

6.7 (U)SAT Framework behaviour

The (U)SAT Framework is a (U)SAT extension of the CAT Runtime Environment as defined in ETSI TS 102 241 [2]. In addition, the (U)SAT Framework shall consider the EVENT_EVENT_DOWNLOAD_* defined in this specification when issuing the SET UP EVENT LIST system proactive command.

7 UICC toolkit applet

See ETSI TS 102 241 [2].

8 Geo Location API

The Geo Location API consists of the package *uicc.usim.geolocation*. This package defines services to allow an Applet to perform a geographical location operation, depending of the ME capabilities. When a geographical location operation is requested, the API will follow a defined way to choose either "Geographical Location Request" toolkit command or "Provide Local Information" toolkit command as defined in TS 31.111 [7] to determine the location information. The result is formatted using GAD shapes as defined in TS 23.032 [14] or in the format of NMEA sentences defined in IEC 61162-1 [15].

9 SUCI API

The SUCI API consists of the package *uicc.usim.suci*. This package defines services to allow an Applet to perform a SUCI computation upon reception of terminal request.

If an applet has registered an object implementing the interface *SUCICalculator* to the USIM application, then when the ME sends a GET IDENTITY APDU Command in SUCI context to this USIM application, the (U)SAT framework shall invoke the *getSUCI* method in order to retrieve the SUCI to be returned to the ME. Only one object can be registered per USIM application. The reference to the object is needed to dereference the object. If no object is registered, the USIM shall return the SUCI computed by its own means, according to the USIM configuration.

If an exception is raised, the (U)SAT framework behaviour is implementation specific.

Annex A (normative): Java Card™ (U)SIM API

The attached files "31130_Annex_A_Java.zip", and "31130_Annex_A_HTML.zip" contains source files and html documentation for the Java Card™ (U)SIM API.

Annex B (normative): Java Card™ (U)SIM API identifiers

The attached file "31130_Annex_B_Export_files.zip" contains the export files for the uicc.usim.* package.

Annex C (normative): (U)SIM API package version management

The following table describes the relationship between each TS 31.130 specification version and its packages AID and Major, Minor versions defined in the export files.

uicc.usim.access package		
TS 31.130	Major, Minor	AID
	1.0	A000000087 1005 FFFF FFFF 89 13 100000
7.7.1	1.1	
8.3.0	1.2	
9.1.0	1.3	
12.0.0	1.4	

uicc.usim.toolkit package		
TS 31.130	Major, Minor	AID
	1.0	A000000087 1005 FFFF FFFF 89 13 200000
7.1.0	1.1	
7.2.1	1.2	
7.7.1	1.3	
7.9.0	1.4	
8.3.0	1.5	
9.1.0	1.6	
9.4.0	1.7	
10.4.0	1.8	
14.2.0	1.9	

uicc.usim.geolocation package		
TS 31.130	Major, Minor	AID
	1.0	A000000087 1005 FFFF FFFF 89 13 300000
13.1.0	2.0	

uicc.usim.suci package		
TS 31.130	Major, Minor	AID
15.1.0	1.0	A000000087 1005 FFFF FFFF 89 13 400000

The package AID coding is defined in ETSI TS 101 220 [1]. The (U)SIM API packages' AID are not modified by changes to Major or Minor Version.

The Major Version shall be incremented if a change to the specification introduces byte code incompatibility with the previous version.

The Minor Version shall be incremented if a change to the specification does not introduce byte code incompatibility with the previous version.

For a table describing the versioning of a package, a line is introduced only upon changes of Major or Minor version of its package.

The package *uicc.usim.access* contains only constants, therefore it may not be loaded on the UICC.

Annex D (normative): USIM API jar files

The attached files "31130_Annex_D.jar", contains class files for the Java Card™ (U)SIM API.

Annex E (informative): Change History

Change history							
Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
	TP-27					Generation of Version 7.0.0 based on version 6.2.0	7.0.0
	TP-27	TP-050023	009			Allow passing of specified status words through the (U)SAT Framework	7.0.0
	CT-28	CP-050139	011			Align paragraph numbering between TS 31.130 and ETSI TS 102 241	7.1.0
	CT-28	CP-050139	013			Delete version and author info from the Java source code	7.1.0
	CT-28	CP-050141	014			Addition of new events EVENT_FORMATTED_USSD and EVENT_UNFORMATTED_USSD	7.1.0
	CT-29	CP-050340	016			Adding missing constant values	7.2.0
						2005-11: Adds missing attachment files and adds line to table in annex C.	7.2.1
	CT-33	CP-060391	019	1		Correction of misnamed constant	7.3.0
			020	1		Addition of missing event download I-WLAN access status	
	CT-34	CP-060546	0022	2		Clarification on getShortMessageLength() method when applied on a SMS Cell Broadcast.	7.4.0
		CP-050548	0024	1		Correction of the USATEnvelopeHandlerSystem method prototype	
	CT-35	CP-070068	0027	1		Correction of Annex A JAVA.zip, package uicc.usim.toolkit	7.5.0
			0028	2		Update the reference to Java Card 2.2.2	
	CT-36	CP-070302	0029	-		Correction of the reference to ETSI TS 102 241	7.6.0
		CP-070298	0029	-		Correction of references to ETSI TS 102 223 and ETSI TS 102 221	
	CT-38	CP-070844	0032	1		Introduction of new constant values for files in the USIM application	7.7.0
						Annex A and B attachments provided (2008-08)	7.7.1
	CT-42	CP-080908	0034	2		Introduction of a geographical location discovery Java Card™ API	8.0.0
	CT-43	CP-090196	0035	1		Introduction of missing constant values for USIM files	8.1.0
	CT-45	CP-090719	0039	2		Alignment of constants with 31.111	8.2.0
	CT-46	CP-090788	0040	1		References update	8.3.0
	CT-46	CP-091013	0042	1		Support of missing event EVENT_EVENT_DOWNLOAD_NETWORK_REJECTION	8.3.0
	CT-46	CP-091013	0045	1		Support of missing constants in USAT Terminal Profile	8.3.0
	CT-46	-	-	-		Upgrade of the specification to Rel-9	9.0.0
	CT-47	CP-100185	0047	1		Addition of missing constant values	9.1.0
	CT-47	CP-100198	0048	2		Supporting operator controlled CSG list for H(e)NB	9.1.0
	CT-47	CP-100198	0049	2		Support of CSG cell discovery and CSG selection event	9.1.0
	-----					Spec reissued as v9.1.1 due to a bad version number on the cover sheet	9.1.1
	CT-50	CP-100836	0046	1		Update reference to "Java Card 3.0.1 Classic" reference	9.2.0
	SP-51	-	-	-		Upgrade of the specification to Rel-10	10.0.0
	CT-52	CP-110507	0050	1		Addition of events and reservation of constant values for Java API	10.1.0
	CT-54	CP-110905	0053	-		Correction to TAG_CSG_SELECTION_STATUS	10.2.0
	CT-54	CP-110905	0054	-		Correction to constant value in TerminalProfile.java	10.2.0
	CT-55	CP-120154	0059			Correction to TAG_CSG_SELECTION_STATUS	10.3.0
	CT-55	CP-120154	0060			Correction to constant value in TerminalProfile.java	10.3.0
	CT-55	CP-120154	0058	1		Update the reference to ETSI TS 102 241	10.3.0
	-----					Editorial version correcting the three lines above	10.3.1
	CT-56	CP-120393	0061	1		Correct implementation of CR 0059 for TAG_CSG_SELECTION_STATUS_N	10.4.0
	CT-56	CP-120392	0062	1		Adding a constant value in USATTerminalProfile.java for the indication of IMS support	10.4.0
	CT-56	CP-120393	0063	1		Adding constant values in USIMConstants.java for missing file identifiers	10.4.0
	SP-57					Automatic upgrade to Rel-11	11.0.0
	SP-65					Automatic upgrade to Rel-12	12.0.0
	CT-70	CP-150827	0071			Missing rule for SMS_PP envelope response handling	13.0.0
	CT-73	CP-160550	0072	5		Geo Location API corrections Note 1: known problem within the change request, to be fixed at CT-74 Note 2: in the CR, the body of the CR and the attached annexes are not identical. The body of the CR contains the correct text and is implemented.	13.1.0
	CT-74	CP-160788	0073	1		Geo Location API format alignment	13.2.0
	CT-75	CP-170166	0075			Geolocalization API document alignment	13.3.0
	SA-75					Automatic upgrade to Rel-14	14.0.0
	CT-78	CP-173150	0077	-		Update of reference to ETSI TS 102 241	14.1.0
	CT-78	CP-173150	0078	-		Editorial change of Java Card reference	14.1.0
	CT-78	CP-173143	0076	3		Corrections in Annex C	15.0.0
						Added missing attachments	15.0.1
2019-03	CT#83	TP-050023	0081	1	F	SUCI Package	15.1.0
2019-09	CT#85	CP-192013	0085	1	F	Add support for ENVELOPE (EVENT DOWNLOAD - Data Connection Status Change)	15.2.0

2019-09	CT#85	CP-192014	0083	1	F	Update of reference to ETSI TS 102 241	15.2.0
2019-09	CT#85	CP-192014	0084	-	F	Clarification for SUCI API	15.2.0
2020-01						5G logo updated in a cover page as agreed in CT#86	15.2.1
2020-02						Attachments updated	15.2.2
2020-06	CT#88e	CP-201148	0086	3	F	Update the scope of 31.130 to cover 4/5G aspects	15.3.0
2020-07	-	-	-	-	-	Update to Rel-16 version (MCC)	16.0.0